INCH-POUND

AN924 Rev 12
w/AMENDMENT 1
23 January 2013
SUPERSEDING
AN924 Rev 12
14 June 2012

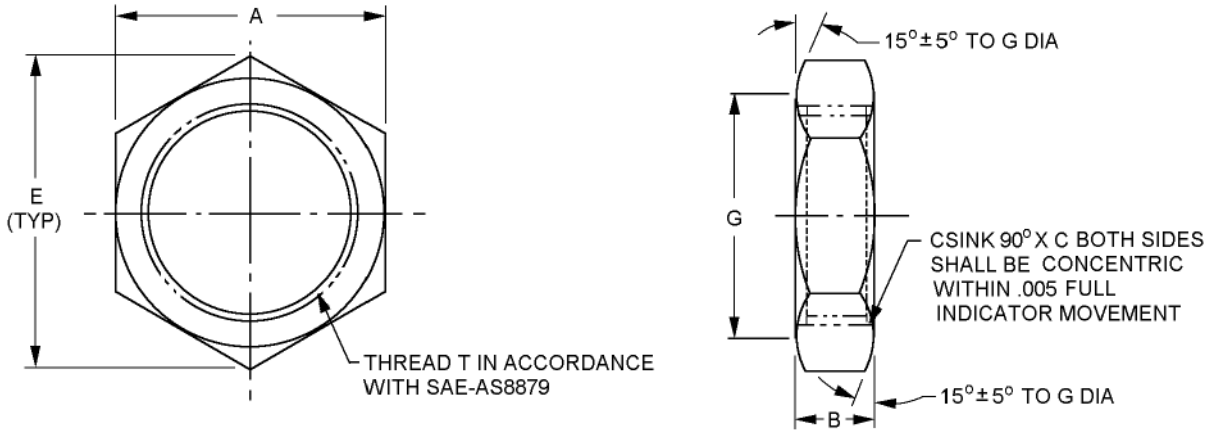
DETAIL SPECIFICATION SHEET

NUT, TUBE, BULKHEAD AND UNIVERSAL FITTING

Reinstated after 14 June 2012. Inactive for new design.
For new design, use SAE-AS5178.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4842.




| Inch | mm |
|---|---|
| .005 | 0.13 |

FIGURE 1. Tube nut dimensions and configurations.

AMSC N/A

FSC 4730

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| Dash number | Tubing OD | Thread T (Ref) SAE-AS8879 | A inches (mm) | | B inches (mm) |
|---|---|---|---|---|---|
| 2 | .125 | .3125-24UNJF-3B | .563 (14.30) | +.003 (0.08) -.004 (0.10) | .219 (5.56) |
| 3 | .188 | .3750-24UNJF-3B | .625 (15.88) | | .219 (5.56) |
| 4 | .250 | .4375-20UNJF-3B | .688 (17.48) | | .250 (6.35) |
| 5 | .313 | .5000-20UNJF-3B | .750 (19.05) | | .250 (6.35) |
| 6 | .375 | .3750-18UNJF-3B | .813 (20.65) | | .266 (6.76) |
| 8 | .500 | .7500-16UNJF-3B | 1.000 (25.40) | | .313 (7.95) |
| 10 | .628 | .8750-14UNJF-3B | 1.125 (28.58) | | .359 (9.12) |
| 12 | .750 | 1.0625-12UNJ-3B | 1.375 (34.93) | | .406 (10.31) |
| 16 | 1.000 | 1.3125-12UNJ-3B | 1.625 (41.28) | ±.016 (0.41) | |
| 20 | 1.250 | 1.6250-12UNJ-3B | 1.938 (49.23) | | |
| 24 | 1.500 | 1.875-12UNJ-3B | 2.188 (55.58) | ±.020 (0.51) | |
| 28 | 1.750 | 2.250-12UNJ-3B | 2.563 (65.10) | | |
| 32 | 2.000 | 2.500-12UNJ-3B | 2.813 (71.45) | | |
| 40 | 2.500 | 3.000-12UNJ-3B | 3.312 (84.12) | | |
| 48 | 3.500 | 3.500-12UNJ-3B | 3.812 (96.82) | | |

| Dash number | C Dia. | E Min. | G ±.010 (0.25) inches (mm) | H Max (see note 5) inches (mm) |
|---|---|---|---|---|
| 2 | .313 (7.95) | .629 (15.98) | .500 (12.70) | .005 (0.13) |
| 3 | .375 (9.53) | .699 (17.75) | .562 (14.27) | |
| 4 | .438 (11.13) | .771 (19.58) | .625 (15.88) | |
| 5 | .500 (12.70) | .842 (21.39) | .687 (17.45) | |
| 6 | .563 (14.30) | .914 (23.22) | .750 (19.05) | |
| 8 | .750 (19.05) | 1.127 (28.63) | .937 (23.80) | |
| 10 | .875(22.23) | 1.270 (32.26) | 1.052 (26.72) | |
| 12 | 1.063 (27.00) | 1.555 (39.50) | 1.274 (32.36) | .008 (0.20) |
| 16 | 1.313 (33.35) | 1.827 (46.41) | 1.524 (38.71) | |
| 20 | 1.625 (41.28) | 2.184 (55.47) | 1.838 (46.69) | |
| 24 | 1.875 (47.63) | 2.465 (62.61) | 2.093 (53.16) | |
| 28 | 2.250 (57.15) | 2.892 (73.46) | 2.468 (62.69) | |
| 32 | 2.500 (63.50) | 3.178 (80.72) | 2.718 (69.04) | |
| 40 | 3.000 (76.20) | 3.747 (95.17) | 3.218 (81.74) | |
| 48 | 3.500 (88.90) | 4.318 (109.68) | 3.718 (94.44) | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.016 inch (0.41 mm) angles ±0.5°.
4. Break sharp edges .and remove all hanging burrs and slivers.
5. Squareness between thread and face of hex shall not exceed H when measured at diameter G, both sides.
6. Machined surfaces shall be finished to 125µin Ra, forged surfaces shall be 250µin Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
7. For design features purposes, this standard takes precedence over documents referenced herein.
8. Referenced documents shall be of the issue in effect on date of invitation for bid.

FIGURE 1. Tube nut dimensions and configurations - Continued.



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REQUIREMENTS:

Dimensions and configuration shall be in accordance with figure 1.

Installation shall be in accordance with MS21344.

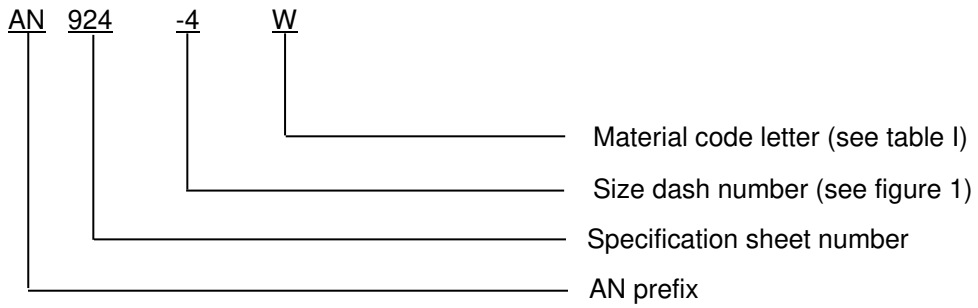
Materials and finishes shall be in accordance with SAE-AS4842, see table I for material code.

TABLE I. Material and code letters.

| Code letter | Material |
|---|---|
| Blank | Steel |
| J | CRES, type 304 |
| K | CRES, type 316 |
| R | CRES, type 321 |
| T 1/ | Titanium |
| W | Aluminum alloy 7075-T73 |

1/ Not for use in oxygen systems.

Part or Identifying Number (PIN): The PIN consists of the letter "AN" the specification sheet number, a dash number for pipe thread size, and a material code letter. Unassigned PIN's shall not be used.




PIN example: AN924-4W indicates a tube nut, .4375-20UNJF-3B threads, aluminum alloy 7075-T73.

Marking: Part shall be impression stamped with the AN PIN, and include the manufacturers CAGE, name, or trademark on the hex side only.

Supersession data:

Due to stress corrosion cracking aluminum alloys 2014 and 2024 "D" designator has been replaced by aluminum alloy 7075 "W" designator example: AN924-4D use AN924-4W.

Metal cracking due to high temperatures CRES alloy 347 "S" designator has been replaced by CRES alloy 321 "R" designator. Example: AN9244S use AN924-4R.

AN924 and MS24400 CRES nuts of like size are physically interchangeable; however MS24400 nuts coded C cannot always replace AN924 nuts coded J or K and MS24400 nuts coded S cannot always replace AN924 nuts coded S.



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Table II provides a detailed cross-reference of AN924 PINs and replacement SAE-AS5178 PINs. Users are cautioned to evaluate replacements for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

TABLE II. AN924 to SAE- AS5178 cross-reference data. 1/

| AN PIN | Tube Size | Replacement AN PIN | AS PIN |
|---|---|---|---|
| AN924-2 | .125 | | AS5178-02 |
| AN924-2D | .125 | AN924-2W | AS5178W02 |
| AN924-2J | .125 | | AS5178J02 |
| AN924-2K | .125 | | AS5178K02 |
| AN924-2S | .125 | AN924-2R | AS5178S02 |
| AN924-2T | .125 | | AS5178T02 |
| AN924-2W | .125 | | AS5178W02 |
| AN924-3 | .188 | | AS5178-03 |
| AN924-3D | .188 | AN924-3W | AS5178W03 |
| AN924-3J | .188 | | AS5178J03 |
| AN924-3K | .188 | | AS5178K03 |
| AN924-3S | .188 | AN924-3R | AS5178S03 |
| AN924-3T | .188 | | AS5178T03 |
| AN924-3W | .188 | | AS5178W03 |
| AN924-4 | .250 | | AS5178-04 |
| AN924-4D | .250 | AN924-4W | AS5178W04 |
| AN924-4J | .250 | | AS5178J04 |
| AN924-4K | .250 | | AS5178K04 |
| AN924-4S | .250 | AN924-4R | AS5178S04 |
| AN924-4T | .250 | | AS5178T04 |
| AN924-4W | .250 | | AS5178W04 |
| AN924-5 | .312 | | AS5178-05 |
| AN924-5D | .312 | AN924-5W | AS5178W05 |
| AN924-5J | .312 | | AS5178J05 |
| AN924-5K | .312 | | AS5178K05 |
| AN924-5S | .312 | AN924-5R | AS5178S05 |
| AN924-5T | .312 | | AS5178T05 |
| AN924-5W | .312 | | AS5178W05 |
| AN924-6 | .375 | | AS5178-06 |
| AN924-6D | .375 | AN924-6W | AS5178W06 |
| AN924-6J | .375 | | AS5178J06 |
| AN924-6K | .375 | | AS5178K06 |
| AN924-6S | .375 | AN924-6R | AS5178S06 |
| AN924-6T | .375 | | AS5178T06 |
| AN924-6W | .375 | | AS5178W06 |

See note at end of table.



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TABLE II. AN924 to SAE- AS5178 cross-reference data - Continued. 1/

| AN PIN | Tube Size | Replacement AN PIN | AS PIN |
|---|---|---|---|
| AN924-8 | .500 | | AS5178-08 |
| AN924-8D | .500 | AN924-8W | AS5178W08 |
| AN924-8J | .500 | | AS5178J08 |
| AN924-8K | .500 | | AS5178K08 |
| AN924-8S | .500 | AN924-8R | AS5178S08 |
| AN924-8T | .500 | | AS5178T08 |
| AN924-8W | .500 | | AS5178W08 |
| | | | |
| AN924-10 | .625 | | AS5178-10 |
| AN924-10D | .625 | AN924-10W | AS5178W10 |
| AN924-10J | .625 | | AS5178J10 |
| AN924-10K | .625 | | AS5178K10 |
| AN924-10S | .625 | AN924-10R | AS5178S10 |
| AN924-10T | .625 | | AS5178T10 |
| AN924-10W | .625 | | AS5178W10 |
| | | | |
| AN924-12 | .750 | | AS5178-12 |
| AN924-12D | .750 | AN924-12W | AS5178W12 |
| AN924-12J | .750 | | AS5178J12 |
| AN924-12K | .750 | | AS5178K12 |
| AN924-12S | .750 | AN924-12R | AS5178S12 |
| AN924-12T | .750 | | AS5178T12 |
| AN924-12W | .750 | | |
| | | | |
| AN924-16 | 1.000 | | AS5178-16 |
| AN924-16D | 1.000 | AN924-16W | AS5178W16 |
| AN924-16J | 1.000 | | AS5178J16 |
| AN924-16K | 1.000 | | AS5178K16 |
| AN924-16S | 1.000 | AN924-16R | AS5178S16 |
| AN924-16T | 1.000 | | AS5178T16 |
| AN924-16W | 1.000 | | AS5178W16 |
| | | | |
| AN924-20 | 1.250 | | AS5178-20 |
| AN924-20D | 1.250 | AN924-20W | AS5178W20 |
| AN924-20J | 1.250 | | AS5178J20 |
| AN924-20K | 1.250 | | AS5178K20 |
| AN924-20S | 1.250 | AN924-20R | AS5178S20 |
| AN924-20T | 1.250 | | AS5178T20 |
| AN924-20W | 1.250 | | AS5178W20 |

See note at end of table.



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TABLE II. AN924 to SAE- AS5178 cross-reference data - Continued. 1/

| AN PIN | Tube Size | Replacement AN PIN | AS PIN |
|---|---|---|---|
| AN924-24 | 1.500 | | AS5178-24 |
| AN924-24D | 1.500 | AN924-24W | AS5178D24 |
| AN924-24J | 1.500 | | AS5178J24 |
| AN924-24K | 1.500 | | AS5178K24 |
| AN924-24S | 1.500 | AN924-24R | AS5178S24 |
| AN924-24T | 1.500 | | AS5178T24 |
| AN924-24W | 1.500 | | AS5178W24 |
| AN924-28 | 1.750 | | AS5178-28 |
| AN924-28D | 1.750 | AN924-28W | AS5178W28 |
| AN924-28J | 1.750 | | AS5178J28 |
| AN924-28K | 1.750 | | AS5178K28 |
| AN924-28S | 1.750 | AN924-28R | AS5178R28 |
| AN924-28T | 1.750 | | AS5178T28 |
| AN924-28W | 1.750 | | AS5178W28 |
| AN924-32 | 2.000 | | AS5178-32 |
| AN924-32D | 2.000 | AN924-32W | AS5178W32 |
| AN924-32J | 2.000 | | AS5178J32 |
| AN924-32K | 2.000 | | AS5178K32 |
| AN924-32S | 2.000 | AN924-32R | AS5178R32 |
| AN924-32T | 2.000 | | AS5178T32 |
| AN924-32W | 2.000 | | AS5178W32 |
| AN924-40 | 2.500 | | AS5178-40 |
| AN924-40D | 2.500 | AN924-40W | AS5178D40 |
| AN924-40J | 2.500 | | AS5178J40 |
| AN924-40K | 2.500 | | AS5178K40 |
| AN924-40S | 2.500 | AN924-40R | AS5178S40 |
| AN924-40T | 2.500 | | AS5178T40 |
| AN924-40W | 2.500 | | AS5178W40 |
| AN924-48 | 3.000 | | AS5178-48 |
| AN924-48D | 3.000 | AN924-48W | AS5178D48 |
| AN924-48J | 3.000 | | AS5178J48 |
| AN924-48K | 3.000 | | AS5178K48 |
| AN924-48S | 3.000 | AN924-48R | AS5178S48 |
| AN924-48T | 3.000 | | AS5178T48 |
| AN924-48W | 3.000 | | AS5178W48 |

1/ For new design use material designator R and W.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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Referenced documents. In addition to SAE-AS4842, this document references the following:

MS21344
ASME B46.1
SAE-AS5178
SAE-AS8879

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2012-100)

Review activities:
Navy - SA
Air Force - 71

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